

Relevance: The Past Can Still Matter in More Than One Way

Human-readable companion to “What Does It Mean for the Past to Still Matter? A Historical-Relevance Audit of Causal Persistence, Information, and Control”

Human-readable companion to the NIP SubV2 manuscript. This companion preserves the scientific claims, limits, open questions, and contraction rules of the source while foregrounding the originating human question: when we say the past “still matters,” what exactly is still present now?

Abstract

Something can have happened earlier and still matter now in several different ways. The earlier event may still causally change a present target. Information about it may be available now, either because a trace survived or because the answer was reconstructed from current cues. That information may change what the system should do under its present objective. And the system may or may not actually use it when acting. Those relations often agree in ordinary life, which is why phrases such as “it still matters,” “I remember,” or “the system knows” sound singular. Relevance argues that they are not singular scientific claims. Its central rule is simple: evidence for one present relation to history does not automatically identify the others.

1. The human problem: “the past still matters” can mean several different things

Imagine saying, “That experience still affects me.” The sentence could mean that the earlier event still changes what happens now. It could mean that you can remember what happened. It could mean that what happened should matter to the decision you are making today. Or it could mean that your present behavior is actually being controlled by that historical information. Those are related possibilities, but they are not interchangeable.

The paper starts from exactly that ambiguity. A past event can leave a physical consequence after the memory is gone. A memory can remain after the external causal consequence has disappeared. A correct historical answer can be reconstructed from current evidence rather than retained from the original event. Information can be available but irrelevant to the current decision, or relevant to the correct decision but ignored by the actual policy.

The paper’s discipline: Do not infer one form of historical relevance from evidence that identifies only another.

2. A present state can be enough for one question and not enough for another

Established science already compresses history. Markov models, belief states, predictive-state representations, and causal abstractions all allow a present description to replace some or all of the literal past when that description is sufficient for the query being asked. Relevance does not challenge that. It asks what happens when several different queries concern the same past distinction.

A present state may be perfectly adequate for predicting what happens next while failing to preserve which earlier intervention still has causal support. A present pattern may let an outside decoder recover a historical feature without showing that the system itself can access or use it. A correct report may come from retention or reconstruction. And a historical fact may be relevant to an optimal decision under one objective but irrelevant under another.

So “history matters” is not treated as an intrinsic property of an event. It is always shorthand for a relation between some historical distinction and a declared present target, information pathway, objective, or policy.

3. The four-question historical-relevance audit

The manuscript reduces the audit to four questions. None is claimed as a new primitive; the contribution is keeping their evidential jobs separate when they are applied to the same history.

Question	Plain-language test	A positive result does not by itself prove
External causal persistence	If the earlier distinction were changed while the relevant alternatives were held fixed, would the present target change?	Memory, access, decision authority, or actual use
Historical information / provenance	Can the system obtain the historical information now, and did it survive from the event or get reconstructed from current cues?	Continuous online maintenance, authority, or actual use
Decision authority	Under the current objective, would changing the historical information change what the system should do?	That the information is accessible or actually used
Realized use	If the present historical signal is changed while competing present cues are fixed, does the actual policy change?	That the resulting action is warranted, valuable, or conscious

Predictive relevance is intentionally not promoted to a fifth core relation. Prediction already has mature frameworks. Here it is an ownership boundary and comparison class: being predictive is useful, but it does not settle the causal, provenance, authority, or use questions.

4. Same endpoint control does not mean the same part of the past still matters

The causal witness uses three earlier binary interventions, A0, A1, and A2. Two systems end with the same number of reachable endpoint states and the same deterministic endpoint capacity. In one system the endpoint retains A0 together with A2; in the other it retains A1 together with A2. From the standpoint of aggregate endpoint control, the channels can look equivalent.

But intervene on A0 while holding the later coordinates fixed and the equivalence disappears. In the first system, changing A0 always changes the endpoint. In the second, changing A0 never changes it because that distinction has reconverged before the endpoint. The bounded conclusion is not a new theory of causation: endpoint causal capacity does not, in general, identify its temporal factorization.

Same overall ability to influence the endpoint does not tell you which earlier distinction still carries that influence.

5. Having a history is not the same as knowing where you are in it

The second constructive case moves from an external endpoint to cognition. Any bounded thought or deliberative episode has an objective duration and a sequence of internal changes. Neither fact proves that the ongoing process possesses a usable elapsed-time coordinate. A later duration judgment can be reconstructed from event structure, remembered content, task regularities, a queried clock, or some combination.

The same case separates access from authority. Precise elapsed time can be available without belonging in the continuation decision. Conversely, time can matter to the task even when precise external time is unavailable. Existing timing research already shows strategic clock monitoring; the narrower Access $\times$ Authority crossing proposed by the paper remains empirically open.

This gives the synthesis a clean chain of non-implications: process history does not establish the queried historical representation; access does not establish decision authority; authority does not establish realized use.

6. A correct historical answer does not tell you where the answer came from

Suppose an earlier fact H and a current cue C normally agree. One system may retain H into a present representation M. Another may have no surviving H-to-M route at all and instead reconstruct the correct answer from C. As long as $H = C$ in ordinary operation, both systems can report the history perfectly.

Accuracy therefore cannot identify provenance. The useful intervention is to break the correlation: vary H while holding the reconstructive route fixed, or otherwise decorrelate H from C. If M follows H, evidence for retained historical provenance strengthens. If M follows C, the ordinary correct answer was reconstructive relative to the tested route.

Reconstruction is not treated as a defect. A reconstruction can be accurate and useful. The only claim is that “the system knows what happened” does not tell us why it can answer correctly now.

7. The matched-present construction shows why ordinary success can hide all four distinctions

The paper then combines the problem in one finite witness. Let an earlier history H and a current cue C normally come from the same source, so ordinary operation gives $H = C$. Add an external target Z, a present historical representation M, a decision D, and four structural switches:

- p: whether Z follows the historical coordinate H or the current cue C;
- r: whether M follows H or is reconstructed from C;
- u: whether D follows the historical representation M or the competing cue C;
- a: whether the current objective rewards following H or following C.

Under the ordinary regime where H and C agree, all 16 possible assignments of those four switches can produce the same endpoint, the same report, the same action, and maximal reward. In other words, ordinary observation can make structurally different systems look identical.

The structures separate only when the redundancy is broken. Put H and C in conflict to audit external persistence or provenance. Hold the world fixed while putting M and C in conflict to test realized use. Change the objective to determine which information should govern the decision. The point is diagnostic: the same present can support different answers to “what from the past is still here?”

8. Each claim needs the intervention aimed at that claim

This is where Relevance becomes more than terminology. It tells the experimenter where to intervene. If the claim is external persistence, manipulate the historical variable and measure the target. If the claim is provenance, decorrelate the original history from reconstructive cues. If the claim is authority, change the objective so the same information should matter in one condition and not another. If the claim is use, perturb the candidate historical representation while fixing the declared competing cues and observe the actual policy.

The paper repeatedly refuses endpoint promotion. A correct endpoint does not identify the historical route. A correct report establishes recoverability more directly than online maintenance. External decodability establishes information for the decoder, not necessarily representation or use by the system. Optimal-policy relevance establishes authority, not implementation. Behavioral sensitivity to a memory channel establishes dependence, not that the action was the history-warranted one.

9. The distinctions are required to collapse when they stop doing scientific work

A framework like this becomes empty if it can always preserve its own distinctions. The manuscript therefore makes contraction part of the theory:

- If the present state is sufficient for every declared predictive, causal, representational, and control query under the relevant interventions, no extra historical variable is scientifically needed for those queries.
- If predictive equivalence also preserves every causal-support query of interest, prediction and causal support need not be separated in that domain.
- If retained and reconstructed information produce the same action under every admissible intervention and provenance is not itself a scientific target, provenance can be ignored.
- If representation and policy use cannot be varied separately in the chosen architecture, that experiment cannot identify them separately. The conclusion is local; it does not prove they are universally identical.
- Temporal factorization matters only when “earlier” and “later” coordinates have scientific meaning. Otherwise preserving the factorization is bookkeeping.
- If every admissible objective either always depends on the history or never does, authority is trivial by construction.

The framework should collapse rather than manufacture distinctions.

10. The point of Relevance

Relevance is not a universal memory theory, a new causal primitive, a new latent variable, or a claim that researchers routinely make every prohibited inference. The causal and predictive machinery is inherited from mature fields. The temporal Access $\times$ Authority experiment is still open and has no participant dataset behind it.

The contribution is connective and methodological. It follows one historical distinction toward four different destinations in the present: an external causal target, a present historical information state and its provenance, an objective-relative decision, and the actual policy. The matched-present construction shows why ordinary success can leave those destinations unresolved, and the audit assigns a different intervention to each claim.

The human question survives intact at the end: something happened before. In what scientifically identifiable sense is it still here now? A useful answer names the target, asks what historical information is available and where it came from, asks whether the current objective gives that information authority, and finally asks whether the actual controller uses it. Evidence for one step is not evidence for the next unless the bridge has itself been identified.

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